

ROAD-SEGMENT RISK CLASSIFICATION USING DECISION TREES AND LIGHTGBM

Research Paper

Abstract

Road traffic risk varies substantially from one road segment to another because of differences in traffic volume, road geometry, surface condition, speed environment, weather, lighting, intersections, pedestrian activity, and historical crash patterns. Conventional safety assessment often relies on manual inspection or aggregated statistics and may not provide a sufficiently consistent way to classify the risk of individual road segments. This study presents a machine-learning framework for classifying road segments into risk categories using Decision Tree and LightGBM models. The proposed workflow treats each road segment as an observation and combines roadway, traffic, environmental, and historical safety attributes to learn the relationship between measurable conditions and a target risk class. Data preprocessing includes cleaning, missing-value treatment, categorical encoding, feature selection, class-distribution analysis, and train-test splitting. A Decision Tree is used as an interpretable baseline because its hierarchical rules can be converted into understandable safety conditions. LightGBM is then applied as a gradient-boosting model capable of learning nonlinear relationships and interactions among heterogeneous road features. Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC where applicable, and computational measures. The comparison is intended not only to identify the stronger classifier but also to examine the trade-off between predictive performance and interpretability. The framework can support transportation agencies in prioritizing road segments for further engineering review, maintenance, traffic management, and targeted safety interventions. The study also discusses data limitations, class imbalance, explainability, generalization, and opportunities for future real-time risk prediction.

Keywords: road safety, road-segment classification, machine learning, Decision Tree, LightGBM, traffic risk, classification, feature importance, intelligent transportation systems

1. Introduction

Road safety is a major transportation concern because crashes are influenced by a combination of roadway design, traffic behavior, vehicle movement, environmental conditions, and human factors. A road network is not equally hazardous at every location. Risk may be concentrated around particular intersections, curves, high-speed corridors, pedestrian crossings, poorly illuminated sections, or road segments with recurring congestion and adverse surface conditions. Consequently, treating an entire road network as a single homogeneous system can hide important local variations in safety conditions.

Road-segment risk classification attempts to address this problem by assigning each segment to a meaningful category such as low, medium, or high risk. Such classification can be formulated as a supervised machine-learning problem when historical observations or engineered safety labels are available. The model receives a set of features describing a road segment and predicts the risk class associated with that segment. This approach can complement conventional road-safety assessment by providing a repeatable data-driven screening mechanism.

The present study focuses on two tree-based methods: Decision Tree and LightGBM. A Decision Tree recursively divides observations according to feature conditions and produces a sequence of decisions leading to a final class. This makes the model relatively easy to interpret. The source paper supplied for adaptation also emphasizes the usefulness of Decision Trees for variable analysis, clear decision paths, and direct interpretation, while noting the danger of overfitting without appropriate control of tree growth.

LightGBM represents a different approach. It builds an ensemble of decision trees sequentially, where later trees focus on correcting errors made by earlier trees. Its boosting framework is particularly suitable for structured tabular data with nonlinear relationships. Comparing the two models therefore provides both an interpretable baseline and a stronger ensemble learner.

The objectives of this work are to identify useful road-risk features, construct a reproducible preprocessing and classification pipeline, train Decision Tree and LightGBM classifiers, evaluate them using multiple metrics, compare their strengths and weaknesses, and develop a practical interpretation of the resulting risk classes.

1.1 Problem Statement

Transportation organizations collect large quantities of roadway and traffic information, yet converting those observations into consistent segment-level risk categories remains challenging. A road segment can simultaneously have moderate traffic volume, poor pavement condition, a high operating speed, several conflict points, and unfavorable environmental exposure. The combined effect of these variables may be nonlinear, making simple threshold-based assessment inadequate.

Another challenge is that safety datasets frequently contain imbalanced risk classes. High-risk segments may represent a smaller portion of the network than low-risk segments. A model that simply predicts the majority class can achieve apparently high accuracy while failing to identify the locations that require attention most urgently. For this reason, the proposed framework considers precision, recall and F1-score in addition to accuracy.

The central problem addressed by this study is therefore: **How can roadway, traffic, environmental, and historical features be used to reliably classify individual road segments into meaningful risk categories, and which of Decision Tree and LightGBM provides the more useful balance between predictive performance and interpretability?**

Research questions

1. Which roadway and traffic variables contribute most strongly to road-segment risk classification?
2. How effectively can a Decision Tree classify road segments into risk categories?
3. Does LightGBM improve classification performance over a single Decision Tree?
4. How does class imbalance affect the evaluation of the models?
5. Can the learned models provide explanations useful for transportation decision-making?

Scope

The work is focused on supervised classification of road segments. It does not claim that a machine-learning prediction alone establishes causality or replaces professional road-safety engineering. Instead, the classifier is considered a screening and prioritization tool. Final intervention decisions should incorporate field inspection, engineering standards, local traffic conditions, and other evidence.

1.2 Contributions of the Study

The study contributes a structured machine-learning approach for segment-level road-risk classification. First, it organizes heterogeneous roadway information into a common analytical representation. Features can include segment length, lane count, speed limit, average traffic volume, heavy-vehicle proportion, road width, curvature, intersection density, lighting condition, pavement quality, weather indicators, pedestrian activity, and historical crash measures. The exact feature set should be adapted to the available dataset rather than assumed to exist in every road database.

Second, the study establishes a preprocessing pipeline. Raw data are checked for missing values, duplicate records, inconsistent units, invalid measurements, and categorical representations. Numerical variables can be imputed or transformed where necessary, while categorical variables can be encoded into machine-readable values. Exploratory analysis is used to inspect distributions, correlations, outliers, and the balance of the target risk classes.

Third, the work compares two models with different characteristics. The Decision Tree provides transparent rules that can be inspected directly. LightGBM provides an ensemble of boosted trees and is expected to capture more complex relationships in structured data. The comparison therefore considers both predictive metrics and practical usability.

Fourth, the study emphasizes multi-metric evaluation. The supplied reference paper similarly argues that accuracy alone is insufficient for classification assessment and discusses precision, recall, false-positive and false-negative behavior, ROC analysis, and F1-score as complementary measures. ■filecite■turn1file2■L228-L256■

Finally, the study identifies research gaps relevant to road-risk prediction, including limited generalization across regions, inconsistent definitions of risk, class imbalance, missing contextual variables, temporal drift, and the need for explainable predictions.

2. Related Work

Machine learning has increasingly been applied to transportation problems involving crash prediction, traffic-state estimation, anomaly detection, road-condition assessment, and safety prioritization. Classification-based approaches are attractive when the desired output is a categorical safety level rather than an exact crash count. In a typical classification setting, historical observations are assigned labels and the model learns a mapping between input features and those labels.

Tree-based methods are particularly useful for roadway datasets because many transportation variables are nonlinear and interact with one another. For example, the safety implication of a high speed limit may depend on road geometry, traffic volume, lane configuration, and intersection frequency. A linear model may require explicit interaction terms, whereas tree-based methods can discover conditional relationships through successive partitions.

Decision Trees have been used widely in classification because they can express decisions as hierarchical conditions. The reference paper notes that Decision Trees perform variable analysis or feature selection during training and can make decision outcomes visible through the tree structure. It also identifies overfitting as a major limitation when tree growth is not appropriately controlled. ■filecite■turn0file0■L195-L213■

Ensemble tree methods extend the basic tree idea by combining multiple trees. Boosting methods build models sequentially and use the errors of earlier learners to guide later learners. LightGBM is a gradient-boosting framework designed for efficient learning on structured datasets. Its efficiency and ability to model nonlinear feature interactions make it a suitable candidate for road-segment risk classification.

Existing transportation studies also demonstrate that evaluation methodology matters. Reported accuracy can be misleading when risk classes are unbalanced. A safety-oriented model should therefore be assessed for its ability to identify high-risk segments, not simply its overall percentage of correct predictions.

2.1 Research Gap

Although machine learning can improve road-safety screening, several limitations remain. First, studies often use different definitions of a high-risk road segment. Some labels may be based on crash frequency, crash severity, expert assessment, or composite indices. Results obtained with one definition cannot automatically be transferred to another.

Second, roadway datasets are heterogeneous. Traffic counts may be measured at different times, weather observations may have missing values, and road-condition surveys may use different scoring systems. Without careful preprocessing, the model may learn measurement artifacts rather than genuine safety patterns.

Third, class imbalance can make a model appear strong while it performs poorly on the minority high-risk class. This is particularly important because the minority class may be the most operationally important category. Evaluation should therefore include class-wise precision, recall, F1-score, and confusion-matrix analysis.

Fourth, predictive performance does not automatically imply practical usefulness. A black-box model may identify high-risk segments but provide little explanation to engineers or administrators. The comparison with a Decision Tree is therefore valuable because it creates an interpretable reference model. Feature-importance analysis and local explanations can further improve the usability of LightGBM.

Finally, road risk changes over time. Construction, resurfacing, traffic growth, changes in speed regulation, seasonal weather, and new intersections can alter the underlying distribution. A model trained on historical data must therefore be monitored and periodically retrained. These gaps motivate a framework that combines robust evaluation, interpretable models, and efficient ensemble learning.

3. Background

Road-segment risk classification can be understood as a pipeline that transforms raw transportation observations into a predicted safety category. A road network is first divided into analysis units, commonly called segments. Each segment receives a set of attributes. These attributes may represent infrastructure, traffic, environment, road users, and historical safety outcomes.

Example feature groups

- Road geometry: segment length, number of lanes, lane width, curvature, grade, median type, shoulder condition.
- Traffic: average daily traffic, peak-hour volume, speed, heavy-vehicle percentage, congestion indicators.
- Infrastructure: pavement condition, signage, markings, lighting, drainage, barriers.
- Network context: intersection density, access-point density, nearby schools or pedestrian generators.
- Environment: rainfall, visibility, temperature, surface condition, time of day.
- Safety history: previous crashes, severity-weighted crash count, near-miss indicators where available.

The target variable can be defined using an existing risk label or by transforming a continuous safety score into ordered classes. For example, a project may define Low, Medium, and High risk using domain-specific thresholds. These thresholds should be documented because changing them changes the learning problem.

A typical system architecture consists of data collection, data cleaning, feature engineering, model training, validation, prediction, evaluation, and reporting. The reference paper similarly organizes its review around background, performance measures, machine-learning methods, comparative analysis, research problems, and future directions.

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3.1 Road-Risk Classification Architecture

The proposed architecture contains seven main stages. Stage 1 collects roadway and contextual information. Stage 2 integrates the data into a segment-level table. Stage 3 performs preprocessing and feature engineering. Stage 4 divides the dataset into training and testing subsets. Stage 5 trains the Decision Tree and LightGBM classifiers. Stage 6 evaluates the models using multiple classification measures. Stage 7 produces a risk-class output and an explanation suitable for prioritization.

Conceptual flow

Data sources → Segment definition → Data cleaning → Feature engineering → Train/Test split → Decision Tree + LightGBM → Evaluation → Risk classification → Feature interpretation → Safety prioritization

The segment definition is important because the unit of analysis influences the learned patterns. Very short segments may contain too few observations, whereas very long segments may combine different roadway environments. A consistent segmentation strategy should therefore be selected before model training.

Data integration must also preserve temporal consistency. If crash history is used as an input feature, information from a future period should not accidentally enter the feature values used to predict an earlier risk label. Such leakage can produce unrealistically high evaluation results.

For operational deployment, the architecture can be extended with a dashboard showing segment location, predicted class, probability or confidence, major contributing variables, and historical context. The output should be interpreted as a prioritization signal rather than a definitive statement that a segment will experience a crash.

3.2 Data Pre-processing

Pre-processing is one of the most important stages because machine-learning performance depends strongly on data quality. The first step is data inspection. Column types, ranges, missing values, duplicate records, and unique values are examined. Numerical variables should use consistent units, while categorical variables should have standardized labels.

Missing values may be handled through median or mean imputation for numerical attributes, and mode or an explicit 'Unknown' category for categorical attributes. The chosen strategy should be fitted using the training data and then applied to the test data to avoid leakage.

Outliers require contextual interpretation. An unusually high traffic count may be a genuine observation rather than an error. Therefore, automatic deletion should not be performed without understanding the measurement process. Invalid values, such as negative lane counts or impossible speed measurements, can be flagged for correction or removal.

Feature engineering may include traffic-density indicators, crash rates normalized by exposure, interaction-related counts, weather categories, and road-condition scores. Feature selection can remove redundant variables and reduce computational complexity. The reference paper also describes feature selection as a dimensionality-reduction step intended to retain informative characteristics while simplifying the representation. ■filecite■turn1file2■L215-L227■

Before training, the target distribution should be examined. If one class dominates, stratified splitting, class weighting, resampling, or threshold adjustment can be considered. Any balancing operation must be applied only to the training portion of the data.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps identify the structure of the road-risk dataset before model development. Descriptive statistics can be used to calculate mean, median, minimum, maximum, and standard deviation for numerical features. Histograms and box plots reveal skewness and potential outliers. Bar charts show the distribution of categorical variables and the number of observations in each risk class.

Correlation analysis can identify highly related numerical variables. For example, road width and lane count may be correlated, while traffic volume and congestion measures may also be related. High correlation does not automatically require removing a variable for tree-based models, but it can complicate interpretation of feature importance.

Group-based analysis is especially useful for transportation data. The average risk label or historical crash measure can be compared across speed-limit categories, lighting conditions, pavement-quality levels, intersection-density groups, and traffic-volume ranges. These comparisons provide an initial understanding of which conditions may be associated with higher risk.

EDA should also be performed with respect to geography and time when those fields are available. A model may otherwise learn regional or temporal patterns that do not generalize. For example, if all high-risk segments in the training set come from one area, location may become a proxy for the target.

The outcome of EDA is not a final causal explanation. It is a diagnostic stage used to understand data quality, identify useful features, discover imbalance, and guide model development.

4. Decision Tree Classifier

A Decision Tree is a supervised learning algorithm that recursively partitions the feature space. At each internal node, the algorithm selects a feature and split condition that improves class separation according to a criterion such as Gini impurity or information gain. The process continues until stopping conditions are reached, producing terminal leaves associated with predicted classes.

Basic process

1. Start with all training observations at the root.
2. Evaluate candidate feature splits.
3. Select the split that provides the greatest reduction in impurity or the greatest information gain.
4. Divide the observations into child nodes.
5. Repeat recursively for each child node.
6. Stop when depth, sample size, impurity, or another stopping condition is reached.
7. Assign the majority class or estimated class probability at each leaf.

For road-risk classification, a resulting rule might conceptually resemble: if traffic volume is high and speed environment is high and intersection density exceeds a threshold, then the segment is more likely to belong to a higher-risk class. The actual thresholds are learned from data and should not be interpreted as universal engineering limits.

The main advantages are interpretability, minimal preprocessing requirements, ability to handle nonlinear relationships, and direct visualization. The main limitation is instability: small changes in training data can change the learned tree. Deep trees can also overfit. The source paper specifically identifies the need for pruning or growth control because Decision Trees can become vulnerable to overfitting. ■filecite■turn0file0■L195-L213■

4.1 Decision Tree Parameters and Pruning

Decision Tree performance depends on hyperparameters controlling tree complexity. Maximum depth limits how many decisions can be made from the root to a leaf. Minimum samples per split and minimum samples per leaf prevent the tree from creating rules based on very small groups. Maximum leaf nodes can also limit model complexity.

Pre-pruning controls the tree while it is being constructed. Examples include restricting maximum depth, requiring a minimum number of samples for a split, or requiring a minimum impurity decrease. These controls can reduce training complexity and improve generalization.

Post-pruning first allows a larger tree to be constructed and then removes branches that contribute little predictive value. Cost-complexity pruning is a common strategy in which tree accuracy is balanced against a penalty for unnecessary complexity.

For the proposed project, several parameter combinations can be evaluated using cross-validation on the training set. The final tree should be selected using validation performance rather than test-set performance. The test set must remain untouched until final evaluation.

Interpretability

A Decision Tree can provide feature importance values and human-readable decision paths. For a transportation agency, this can be valuable because a predicted high-risk segment can be accompanied by the conditions that led to the classification. However, the tree's rules describe patterns in the dataset and should not automatically be treated as causal safety laws.

5. LightGBM Classifier

LightGBM is a gradient-boosting framework based on decision trees. Instead of constructing one large tree, boosting builds a sequence of trees. Each new tree attempts to improve the overall model by focusing on residual errors or difficult observations from the current ensemble. The final prediction is obtained by combining the contributions of the individual trees.

This structure is useful for road-risk data because relationships can be nonlinear and interactions can be complex. For example, the relationship between speed and risk may change when traffic density, curvature, or lighting is different. A boosted tree ensemble can model these conditional patterns without requiring the analyst to specify every interaction manually.

Important LightGBM parameters include the number of estimators, learning rate, maximum tree complexity, number of leaves, minimum data required in a leaf, feature and row subsampling controls, and regularization parameters. Increasing model complexity may improve training performance but can also increase overfitting.

A practical training strategy is to begin with a baseline configuration and then use cross-validation or a validation set for tuning. Early stopping can be used when a validation metric stops improving. Class weighting can be considered when high-risk observations are underrepresented.

LightGBM is particularly appropriate for tabular datasets because it can learn from mixed numerical and encoded categorical information and can represent nonlinear interactions efficiently. Nevertheless, its greater predictive power may come at the cost of interpretability compared with a single Decision Tree, so explanation techniques remain important.

5.1 LightGBM Learning Procedure

The LightGBM learning procedure begins with an initial prediction. The training observations are evaluated to determine where the current model makes errors. A new decision tree is constructed to reduce the objective function, and its contribution is added to the existing ensemble. This process is repeated for a predefined number of iterations or until validation performance stops improving.

For multiclass road-risk classification, the model learns a score for each class and converts these scores into class probabilities. The class with the highest probability can be selected as the final prediction. The probability output can also support ranking: segments with high predicted probability of the high-risk class can be prioritized for review.

Model tuning should focus on generalization. A very large number of leaves, excessive boosting rounds, or weak regularization can allow the ensemble to memorize training patterns. Conversely, a model that is too simple may fail to capture meaningful interactions. Validation curves and cross-validation results can help identify an appropriate balance.

Feature importance can be measured using split-based or gain-based methods, but these should be interpreted carefully when features are correlated. Permutation importance or SHAP-based explanations can provide additional evidence about which features influence predictions. For safety applications, explanations should be presented alongside uncertainty and data-quality information.

6. Dataset Design

A road-segment risk dataset should contain one row for each analysis segment and columns representing measurable characteristics. A suitable dataset may be collected from open transportation repositories, municipal road inventories, crash databases, traffic sensors, weather sources, or a project-specific survey. The selected dataset must contain enough observations and a clearly defined target label for supervised learning.

Illustrative dataset schema

Feature	Type	Example meaning
segment_id	Identifier	Unique road segment
traffic_volume	Numeric	Average vehicles per day
speed_limit	Numeric	Posted speed
lanes	Numeric	Number of lanes
intersection_density	Numeric	Intersections per unit length
road_width	Numeric	Approximate carriageway width
curvature	Numeric	Geometry indicator
pavement_condition	Categorical/Numeric	Surface quality
lighting	Categorical	Lighting availability
weather	Categorical/Numeric	Environmental condition
heavy_vehicle_ratio	Numeric	Share of heavy vehicles
historical_crashes	Numeric	Past crash measure
risk_class	Target	Low/Medium/High

The table is illustrative rather than a claim that every dataset contains these exact fields. The project should use the variables actually available in the chosen dataset and document their definitions. If the target is created from crash counts or another continuous measure, the risk thresholds must be specified explicitly.

6.1 Target Variable Construction

Risk classification requires a defensible target variable. If a dataset already provides a risk category, that label can be used after checking its meaning and consistency. If only continuous safety information is available, categories can be created using documented thresholds or domain-informed quantiles.

One possible project formulation is three-class classification: Low Risk, Medium Risk, and High Risk. The boundaries should reflect the project's chosen risk indicator. For example, a composite index could combine normalized crash frequency, crash severity, and traffic exposure. However, arbitrary thresholds should be avoided because they can introduce artificial class differences.

An alternative is to define risk from historical crash outcomes. The analyst may calculate an exposure-adjusted crash rate and then divide segments into ordered categories. Exposure adjustment is important because a heavily traveled road naturally has more opportunities for incidents than a lightly traveled road. The exact formula should be determined from the available variables and transportation domain requirements.

Target construction is one of the most consequential decisions in the study. A highly accurate model can still be operationally weak if the labels are inconsistent or do not represent the intended safety concept. The project should therefore describe the label-generation process in detail and, where possible, perform sensitivity analysis using alternative reasonable thresholds.

If future crash outcomes are used as labels, the feature window must end before the outcome window. This temporal separation prevents future information from leaking into the predictors.

7. Model Training Methodology

The training methodology begins with a reproducible split of the dataset into training and testing subsets. Stratified sampling is preferred for classification because it preserves approximate class proportions in both sets. A common project configuration is 80% training and 20% testing, although the final proportion can be adjusted according to dataset size.

The training data are used for preprocessing parameter estimation, model fitting, hyperparameter tuning, and cross-validation. The test set is reserved for the final unbiased comparison. If repeated experiments are required, a fixed random seed should be recorded to improve reproducibility.

The Decision Tree baseline can be tuned using maximum depth, minimum samples per split, minimum samples per leaf, maximum leaf nodes, splitting criterion, and cost-complexity pruning. LightGBM can be tuned using learning rate, number of estimators, number of leaves, maximum depth where appropriate, subsampling parameters, and regularization.

Cross-validation provides a more reliable estimate of model performance during tuning. Stratified k-fold cross-validation can be used so each fold retains a representative class distribution. Hyperparameters should be selected using validation performance, not the final test set.

The final models are then retrained on the complete training data using the selected parameters and evaluated once on the held-out test data. The resulting metrics and confusion matrices form the basis for model comparison.

7.1 Model Evaluation Metrics

Accuracy is the proportion of all test observations classified correctly. It is easy to understand but can be misleading when classes are imbalanced. Precision measures the proportion of observations predicted as a class that truly belong to that class. Recall measures how many observations of a class are successfully identified. F1-score combines precision and recall through their harmonic mean.

For high-risk road classification, recall can be especially important because failing to identify a genuinely high-risk segment may prevent it from receiving additional review. Precision also matters because excessive false alarms can consume limited inspection and intervention resources. The appropriate balance depends on the operational objective.

A confusion matrix presents actual classes against predicted classes. It allows the analyst to identify whether errors are concentrated between adjacent categories, such as Medium versus High, or whether the model frequently predicts Low for genuinely High-risk segments.

ROC-AUC can be used for binary classification or extended to multiclass settings using one-vs-rest or other aggregation methods. The supplied reference paper explains ROC analysis as a plot of true-positive rate against false-positive rate across thresholds and also emphasizes precision and recall as useful evaluation measures.

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For a transportation application, model evaluation should also consider computation time, stability across folds, calibration of predicted probabilities where relevant, and practical interpretability.

7.2 Comparative Evaluation

The central comparison is between a single Decision Tree and the LightGBM ensemble. The Decision Tree is expected to provide a transparent model whose structure can be visualized and converted into rules. LightGBM is expected to provide greater flexibility by combining many weak learners into a stronger ensemble.

Criterion	Decision Tree	LightGBM
Interpretability	High	Moderate; requires explanation tools
Nonlinear relationships	Good	Very good
Feature interactions	Good	Very good
Overfitting risk	High for deep trees	Controlled through regularization/tuning
Training complexity	Low to moderate	Moderate
Prediction speed	Fast	Fast after training
Rule visualization	Direct	Not direct
Tabular-data performance	Strong baseline	Often strong ensemble choice

The table summarizes expected methodological characteristics, not measured results. Actual performance must be calculated from the selected dataset. If LightGBM obtains a higher F1-score and recall for the High-risk class while maintaining acceptable precision, it would be a strong candidate for deployment. If the performance difference is small, the simpler Decision Tree may be preferable where transparency is the dominant requirement.

The final comparison should report the same test set, same target definition, and same evaluation metrics for both models. This prevents an apparent improvement from being caused by different experimental conditions.

8. Feature Importance and Explainability

Feature importance analysis helps identify which road characteristics contribute most strongly to model decisions. In a Decision Tree, importance can be derived from impurity reduction accumulated across splits. In LightGBM, importance may be calculated using split frequency or gain. These measures are useful for ranking variables but should not be interpreted as proof of causation.

Suppose traffic volume, speed environment, intersection density, pavement condition, and historical crashes emerge as highly influential features. This would indicate that the model relies heavily on these variables within the dataset. It would not mean that changing one variable alone will necessarily reduce crash risk because features may be correlated and the model may use them as proxies for other conditions.

Explainability can be strengthened with local explanations. For a particular segment, an explanation method can show which features pushed the prediction toward Low, Medium, or High risk. This is particularly useful for reviewing borderline cases or communicating results to non-technical stakeholders.

The Decision Tree itself can serve as an interpretable benchmark. The source material highlights the advantage of tree models in exposing possible decision paths and reducing ambiguity in classification decisions.

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A responsible reporting format should display the predicted risk class together with confidence/probability, key contributing features, data timestamp, and a note that the result is a model-based screening output rather than a definitive engineering diagnosis.

8.1 Handling Class Imbalance

Road-risk datasets may contain many more low-risk segments than high-risk segments. If the model optimizes overall accuracy without considering this imbalance, it may learn to favor the majority class. This is unacceptable when the primary objective is to identify segments that need safety attention.

Several approaches can be considered. Stratified sampling ensures that training and evaluation sets contain representative class proportions. Class weights can increase the penalty for errors on minority classes. Oversampling methods can create a more balanced training distribution, while undersampling can reduce the dominance of the majority class. Threshold tuning can also shift the decision boundary when probability outputs are available.

Any resampling operation must be performed inside the training process or within each cross-validation fold. Applying oversampling before splitting can cause duplicated observations to appear in both training and testing sets, leading to leakage and inflated performance.

The preferred strategy should be selected based on validation results. A model with slightly lower accuracy but substantially higher recall for High-risk segments may be more valuable than a model with higher overall accuracy and poor minority-class detection.

The final report should provide class-wise metrics and the confusion matrix so readers can see whether the model is genuinely identifying the risk categories or simply benefiting from class frequency.

9. Experimental Workflow

The complete experiment can be implemented in Python using pandas for data manipulation, NumPy for numerical operations, scikit-learn for preprocessing and Decision Tree modelling, LightGBM for gradient boosting, and Matplotlib or similar libraries for visualization. A notebook environment such as Google Colab can be used when the project requires a cloud-based execution environment.

Workflow

1. Load the road-segment dataset.
2. Inspect dimensions, data types, missing values, and duplicate rows.
3. Define the target risk class.
4. Remove identifiers that would leak the target.
5. Separate predictors and target.
6. Perform train-test splitting with stratification.
7. Fit preprocessing transformations using training data only.
8. Train a baseline Decision Tree.
9. Tune Decision Tree hyperparameters.
10. Train and tune LightGBM.
11. Generate predictions and class probabilities.
12. Calculate accuracy, precision, recall, F1-score and confusion matrix.
13. Compare feature importance and model complexity.
14. Interpret the final results and limitations.

Reproducibility should be maintained by recording the dataset version, feature definitions, random seed, preprocessing choices, model parameters, software versions, and evaluation procedure. This allows the experiment to be repeated after the dataset or model is updated.

9.1 Pseudocode

Algorithm 1: Road-Segment Risk Classification

Input: Road-segment dataset D

Output: Predicted risk class for each test segment

1. Read D.
2. Validate feature types and remove invalid records.
3. Construct or load risk_class.
4. Separate X (features) and y (risk class).
5. Split X,y into stratified training and testing sets.
6. Fit preprocessing transformations on training data.
7. Transform training and testing data.
8. Train Decision Tree DT on training data.
9. Tune DT using cross-validation.
10. Train LightGBM LGBM on training data.
11. Tune LGBM using cross-validation.
12. Predict y_DT and y_LGBM for the test set.
13. Compute accuracy, precision, recall, F1-score and confusion matrix for both models.
14. Compare class-wise performance.
15. Calculate feature importance/explanations.
16. Select the model according to predictive and operational requirements.
17. Return predicted risk class and supporting explanation.

This algorithm follows the same general research logic used in the supplied reference material: data are prepared, machine-learning techniques are applied, and performance is compared using explicit evaluation measures. The reference paper also uses algorithm descriptions and comparative tables to make the methodology reproducible.

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10. Expected Results and Interpretation

The expected outcome is a model capable of separating road segments into meaningful risk categories. Because actual performance depends on the selected dataset, this paper does not invent numerical accuracy or F1-score values. Instead, the experimental results should be filled with values produced by running the models on the project dataset.

Recommended results table

Metric	Decision Tree	LightGBM
Accuracy	Insert measured value	Insert measured value
Precision – High Risk	Insert measured value	Insert measured value
Recall – High Risk	Insert measured value	Insert measured value
F1-score – High Risk	Insert measured value	Insert measured value
Macro F1	Insert measured value	Insert measured value
Weighted F1	Insert measured value	Insert measured value
ROC-AUC	Insert if applicable	Insert if applicable

If LightGBM produces better macro F1 and High-risk recall, it may be considered the stronger predictive model. If the Decision Tree provides similar performance with a much simpler structure, it may remain attractive for operational use. The correct conclusion must be based on measured results rather than an assumed winner.

10.1 Confusion Matrix Analysis

The confusion matrix should be examined after calculating the overall metrics. For three classes, it can reveal whether Low-risk segments are being confused with Medium-risk segments, or whether High-risk segments are frequently classified as Medium or Low. The latter errors may be more important from a safety-management perspective.

For example, if the High-risk recall is low, the model is failing to identify a substantial portion of genuinely high-risk segments. Possible causes include insufficient high-risk training examples, noisy labels, missing explanatory variables, or an overly conservative decision threshold. If precision is low, the model may flag too many segments as high risk, increasing the workload for manual review.

Comparing confusion matrices between Decision Tree and LightGBM can reveal differences hidden by overall accuracy. A LightGBM model may achieve only a modest accuracy improvement but a much larger increase in recall for the High-risk class. Such an improvement can be operationally significant.

The report should therefore present both confusion matrices and class-wise metrics. The supplied reference paper similarly warns that accuracy alone can be insufficient and discusses the importance of false-positive and false-negative behavior. ■filecite■turn1file2■L235-L256■

11. Advantages of the Proposed Approach

The proposed framework offers several advantages. First, it transforms heterogeneous road information into a consistent classification task that can be repeated across a network. Second, the Decision Tree provides a transparent baseline that is easier to communicate to engineers, administrators, and students. Third, LightGBM can model complex nonlinear relationships and interactions without requiring a very large manually specified rule set.

Fourth, multi-metric evaluation provides a more complete view of model behavior. Rather than reporting only one number, the framework examines accuracy, precision, recall, F1-score, confusion matrices and, where appropriate, ROC-AUC. This is important when the costs of different classification errors are unequal.

Fifth, feature importance and explanation techniques can convert model output into information that supports investigation. A segment predicted as high risk can be accompanied by the variables most strongly associated with that prediction, helping analysts decide whether the result warrants field inspection.

Sixth, the framework is scalable. Once a validated model and preprocessing pipeline are available, new segment records can be scored automatically. This creates an opportunity for periodic network-wide screening rather than relying entirely on manual review.

Finally, the framework can be extended. Additional models, temporal features, spatial context, real-time sensor information, and explainable-AI techniques can be incorporated without changing the overall architecture.

11.1 Limitations

The first limitation is data quality. Machine-learning models cannot recover information that is absent from the dataset. If important factors such as driver behavior, actual operating speed, visibility, pedestrian exposure, or temporary road conditions are not recorded, the model may have limited predictive capability.

The second limitation is label quality. A risk class created from historical crashes reflects the way the label was defined and may inherit reporting biases. Crash records may be incomplete, and rare severe events may be underrepresented.

The third limitation is generalization. A model trained on one city, region, climate, or road network may not perform equally well elsewhere because roadway design, traffic behavior, enforcement, and data collection practices differ.

The fourth limitation is temporal drift. Road infrastructure and traffic conditions change over time. A model that performs well on historical data should therefore be monitored after deployment.

The fifth limitation is interpretability of ensemble models. Although LightGBM can provide feature importance, its predictions are not as directly understandable as a single Decision Tree. Explanations should therefore be incorporated into any practical deployment.

These limitations do not invalidate the approach; they define the conditions under which its predictions should be used. The model should support, not replace, professional safety assessment.

12. Future Scope

Future work can extend the framework in several directions. First, temporal modelling can incorporate changing traffic volume, seasonal weather, construction periods, and rolling crash histories. Instead of treating every observation as static, the system could estimate risk for future time windows.

Second, spatial information can be incorporated. Neighboring segments may influence one another because traffic flows through connected networks. Graph-based approaches could model this connectivity, while geospatial features could capture proximity to intersections, schools, hospitals, transit stops, and other activity centers.

Third, real-time data sources can be integrated. Connected traffic signals, road sensors, weather stations, and intelligent transportation systems can provide dynamic information about congestion, visibility, surface conditions, and traffic speed. This would support dynamic rather than purely historical risk classification.

Fourth, explainable machine learning can be expanded. SHAP values, partial-dependence analysis, and local explanations can help communicate LightGBM predictions. A future decision-support dashboard could show predicted risk, confidence, dominant contributing factors, historical trend, and recommended priority level.

Fifth, model ensembles can be investigated. Random Forest, XGBoost, CatBoost, neural networks, and calibrated stacking methods could be compared with Decision Tree and LightGBM. However, additional complexity should only be introduced when it provides measurable value.

Finally, prospective validation is important. A model should be tested on future observations or a geographically separate road network before being relied upon for operational safety decisions.

12.1 Practical Deployment

A practical deployment can be designed as a batch or web-based decision-support system. A scheduled process receives updated road-segment data, applies the stored preprocessing pipeline, generates predictions using the selected model, and writes the results to a database. A dashboard can then display the network with segments grouped by predicted risk class.

For each segment, the interface can show segment identifier, predicted risk class, probability of each class, recent data timestamp, key features, feature-explanation summary, and links to supporting inspection records. A filter can allow users to view only high-risk segments or segments whose risk increased since the previous assessment.

Operational governance should include data validation, model versioning, access control, monitoring, and periodic retraining. Every prediction should be traceable to the model version and data snapshot that generated it.

A human-in-the-loop design is recommended. Transportation professionals should review high-priority predictions before major interventions are selected. Feedback from field inspections can later be incorporated into future training data, provided that the feedback process is carefully documented to avoid introducing new bias.

The system can therefore function as a prioritization layer between raw transportation data and engineering decision-making. It can reduce the amount of manual screening required while keeping final responsibility with qualified personnel.

13. Discussion

The comparison between Decision Tree and LightGBM illustrates an important principle in applied machine learning: the best model is not necessarily the one with the highest single metric. A transportation organization may value transparency, auditability, ease of maintenance, computational efficiency, and consistency in addition to predictive accuracy.

Decision Tree offers a clear representation of learned conditions and can be useful for teaching, exploratory analysis, and communicating a simple screening policy. Its main challenge is controlling complexity. LightGBM offers a more flexible ensemble representation and may perform better when risk depends on many interacting variables. Its main challenge is explaining the ensemble to stakeholders.

The most useful evaluation is therefore multidimensional. Predictive metrics answer whether the model classifies observations correctly. Confusion matrices answer what types of mistakes occur. Feature analysis answers which variables influence the predictions. Validation across time or geography answers whether the model generalizes. Operational review answers whether the predictions are actionable.

This perspective is consistent with the supplied reference paper's broader methodological lesson that machine-learning systems should not be judged by accuracy alone and that comparative evaluation should consider several performance measures and limitations. ■filecite■turn1file2■L228-L256■

For the road-risk problem, the most important practical objective is reliable identification and prioritization of segments requiring additional attention. A model that achieves strong overall accuracy but misses many high-risk segments would not satisfy that objective.

14. Conclusion

This study presented a machine-learning framework for ROAD-SEGMENT RISK CLASSIFICATION USING DECISION TREES AND LIGHTGBM. The approach models each road segment as a structured observation containing roadway, traffic, environmental, infrastructure, and historical safety characteristics. After data cleaning and feature engineering, the dataset is divided into training and testing subsets, and two tree-based classifiers are developed.

The Decision Tree serves as an interpretable baseline. Its hierarchical structure allows analysts to inspect the conditions associated with different risk classes and provides a useful mechanism for communicating model behavior. However, tree depth and pruning must be controlled because an overly complex tree can overfit training data.

LightGBM extends tree-based learning through gradient boosting. By combining many decision trees sequentially, it can capture nonlinear relationships and feature interactions that may be difficult for a single tree to represent. Its suitability for structured tabular data makes it a strong candidate for the road-risk classification task, although its predictions require additional explanation for transparent deployment.

The study recommends evaluating both models using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC where appropriate. Particular attention should be given to recall and F1-score for the High-risk class because failing to identify a genuinely high-risk segment may have greater operational consequences than a false alarm.

The proposed system should be viewed as a decision-support and prioritization mechanism rather than a replacement for road-safety engineering. Future improvements can incorporate temporal, spatial, real-time, and explainable-machine-learning information. With reliable data, carefully defined labels, rigorous validation, and responsible deployment, the framework can provide a useful foundation for intelligent road-safety management.

15. Recommended Project Implementation

For a student implementation, the project can be executed in a notebook environment using a CSV road-risk dataset. The first notebook section should load and inspect the dataset. The second should perform EDA and visualize class distributions. The third should prepare features and target labels. The fourth should train and tune the Decision Tree. The fifth should train and tune LightGBM. The sixth should compare metrics and confusion matrices. The final section should visualize feature importance and summarize findings.

Suggested software requirements

Python 3.x; Google Colab or Jupyter Notebook; pandas; NumPy; scikit-learn; LightGBM; Matplotlib; Seaborn if desired for EDA; and joblib/pickle for saving trained models. A geographic visualization library may be added if latitude/longitude or road geometry is available.

Suggested hardware requirements

A normal student laptop or cloud notebook is sufficient for a moderate-sized tabular dataset. More computational resources may be useful when the dataset contains millions of observations or when extensive hyperparameter tuning is performed.

Expected project outputs

- Cleaned road-segment dataset.
- EDA plots and class-distribution analysis.
- Trained Decision Tree model.
- Trained LightGBM model.
- Confusion matrices for both models.
- Accuracy, precision, recall and F1-score comparison.
- Feature-importance plots.
- Final predicted risk class for each test segment.
- Discussion of limitations and future scope.

16. References

The following references are suitable starting points for the project report. They should be checked against the exact dataset, software versions, and additional literature used in the final implementation.

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Note: Do not present any invented model-performance values as experimental results. Run the Decision Tree and LightGBM models on the selected road-risk dataset and replace the placeholder result fields with the measured values.